Statistical Analysis Plan (SAP)

Full/ Long title of the Trial: Knowledge support for optimising antibiotic prescribing for common infections in general practices: evaluation of the effectiveness of periodic feedback, decision support during consultations and peer comparisons in a cluster randomised trial

Short Study title/ Acronym: BRIT2

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ABBREVIATIONS

AB	Antibiotic
AMR	Antimicrobial Resistance
ANCOVA	Analysis of Covariance
BMI	Body Mass Index
BRIT	Building Rapid Interventions to reduce AMR and over-prescribing of antibiotics
CACE	Complier Average Causal Effect
CCI	Charlson Comorbidity Index
CI	Confidence Interval
CONSORT	Consolidated Standards of Reporting Trials
CPRD	Clinical Practice Research Datalink
cRCT	Cluster Randomised Controlled Trial
EHR	Electronic Healthcare Record
GP	General Practice or General Practitioner
IMD	Index of Multiple Deprivation
IQR	Inter-Quartile Range
KSS	Knowledge Support System
LHS	Learning Healthcare System
NHS	National Health Service
RTI	Respiratory Tract Infection
SAP	Statistical Analysis Plan
SD	Standard Deviation
UTI	Urinary Tract Infection

1 INTRODUCTION

1.1 Background and rationale

Translating research findings into daily clinical practice is a major challenge. There is considerable need for clinical decisions to be based on the best available evidence, but often this evidence is incomplete as, for example, patients with complex multimorbidity or polypharmacy are excluded from trials. The Learning Healthcare System (LHS) approach has been proposed to better integrate research into clinical practice [2]. It involves iterative phases including data analytics (data to knowledge), feedback to clinicians (knowledge to performance) and implementation of quality improvement activities by the clinicians (performance to data) [3]. The cycle of the LHS starts again by evaluating the effectiveness of these quality improvement activities. The analytics phase includes a detailed data analysis of the opportunities and challenges in current clinical practice and the local site (such as analysis of the effectiveness of current activities). The analyses may identify practices that could potentially improve clinical care. The second phase involves review by the clinicians of these results and decide which have sufficient credibility to generate recommendations for change, ideally customised to its own specific circumstances. The third phase involves clinicians implementing these recommendations [3].

The LHS approach can be used in numerous clinical areas. Primary care accounts for 81% of antibiotic prescribing in England. Overuse of antibiotics is a major public health concern as it could lead to antimicrobial resistance. The NHS 2019 Long Term plan states as priority to continue to optimise use, reduce the need for and unintentional exposure to antibiotics. The 5-year Government plan aims to reduce UK antimicrobial use in humans by 15% by 2024. In this project, the LHS approach will be applied to antibiotic prescribing for common infections via a practice + individual prescriber feedback via dashboards and a Knowledge Support System (KSS) designed for use by GPs in patient consultations for common infections.

This KSS will consist of a display on the GP's computer while seeing the ill patient; the GP will need to enter answers to standard questions about the patient's condition and then presented with information on, for example, the patient's risk of complications due to the infection. We will also develop individualised patient leaflets that the GP can provide to patients during consultation after KSS activation.

The overall research question will be to evaluate whether the LHS approach (i.e., detailed data analysis followed by feedback to clinicians via the KSS) improves antibiotic prescribing without increasing the risks of complications relative to the more basic generic feedback to practices and individual GPs outside individual patient consultations via the dashboard.

1.2 Objectives

The specific research question will be to evaluate the effectiveness of the KSS (in a cluster-randomised controlled trial). The results of these analyses will then be used to inform the clinicians about their practice (through regularly updated dashboards) as well as to inform the content of information provided in a KSS during consultation.

The primary outcome will be the overall rate of antibiotic prescribing. One secondary outcome will be the safety outcome of infection-related complications, such as pneumonia and lower respiratory tract infections (RTIs), peritonsillar abscess, mastoiditis, intracranial abscess, empyema, scarlet fever, pyelonephritis, septic arthritis, osteomyelitis, meningitis, toxic shock syndrome and septicaemia, and Lemierre syndrome (as defined by Gulliford et al [4]), within 30 days of initial GP consultation for a common infection as recorded in the primary care records. Another secondary outcome will be the level of risk-based prescribing. For each patient who has been prescribed an antibiotic and who has a recent EHR record of a common infection, the predicted risk of hospitalisation for infection-related complications will be estimated as based on the risk prediction scores as developed and validated by Fahmi et al 2024 [5]. It is expected that any risk-based prescribing by GPs would lead to an increased mean predicted risk in the intervention arm. The final secondary outcome was to have been another safety outcome, namely the rate of hospital admission due to infection-related complications within 30 days of initial GP consultation for a common infection. We considered two potential sources to obtain these data: hospital records and the practice records. However, we have been unable to obtain access to the hospital records for this project and, on inspection of the general practice records available through Graphnet, it was ascertained that the reasons for hospital admission were so poorly recorded that using practice records that trying to identify which admissions were due to infection-related complications rather than for some other reason was impractical. It was therefore decided by the Research Team not to extract this information from Graphnet and hence to drop this secondary outcome variable from the analysis. We therefore do not refer further to the analysis of hospital admission due to infection-related complication in this Statistical Analysis Plan (SAP). The primary and remaining safety secondary outcomes are similar to a recent cRCT of generic decision support with a leaflet in UK primary care [6].

2 TRIAL METHODS

2.1 Trial design

The study is a pragmatic cluster-randomised controlled trial (cRCT) with 1:1 randomisation of practices to:

1. Control: Standard care + prescriber feedback dashboards to provide periodic feedback to general practices ('Enhanced care').
   or
2. Intervention: Prescriber feedback dashboards + KSS during consultation with the clinician providing content (such as an information sheet) that is personalised to the patient ('BRIT2 Intervention').

The study sites will include general practices located in the North-West of England that provide patient-level data to a regional shared care record system 'Graphnet' (https://www.graphnethealth.com), which is deployed in Greater Manchester, Wirral and Cheshire/Merseyside regions, together covering a 5.4 million general population. The shared care record collates data from various NHS organisations to support direct patient care, service planning and research. Anonymised patient-level data corresponding to baseline characteristics and outcomes for this trial will be extracted from Graphnet and aggregated at the practice level for Greater Manchester and Wirral/Cheshire/Merseyside separately. These two aggregated data sets will then be combined to form the trial data set for statistical analyses.

The patients, practice staff and research team will not be blinded to intervention group allocation.

2.2 Randomisation

Allocation to interventions used anonymised practice identifiers with (cluster) randomisation stratified by region (Greater Manchester; Wirral; Merseyside/Cheshire) and the baseline median rate of antibiotic prescribing (high: above England practice median*; low ≤ England practice median*). Practice AB rates were calculated using data from 'OpenPrescribing' (https://openprescribing.net/analyse/). Data from 4 calendar months: June 2022, September 2022, December 2022 and March 2023 were used to estimate the average annual rate for each practice, which was then used to classify the practice as a 'high' or 'low' antibiotic prescribing practice. The cluster randomisation was implemented using Sealed Envelope randomisation system (https://www.sealedenvelope.com). An independent statistician not otherwise involved with the project set up the randomisation system to deliver allocations according to the algorithm specified above. The project manager recruited practices and randomly allocated practices to the intervention groups using the system once eligibility was confirmed.

* The England general practice median used was that for May 2023, which was 46.64 AB prescriptions per 1000 registered adult patients per month.

2.3 Sample size

The number of practices allocated to the KSS arm target was limited to approximately 62 practices as this arm involves time and effort of clinicians during consultation and was therefore restricted for this trial due to treatment resource costs.

This final sample size specification, which was firstly produced for the published protocol paper [1], has been computed based on the primary outcome of practice rate of antibiotic prescribing (per 1000 patients). In CPRD, the mean practice list size was 7981 and the overall rate of the number of antibiotic prescriptions (for any indication) was 598 per 1000 patients per year (SD 155). Assuming a reduction of 10% in the enhanced care (control) arm, its antibiotic prescribing rate will be around 538 per 1000 patients per year. Assuming an unchanged SD of 155 and practice attrition rate of 5%, randomising 62 practices to each of the two arms will provide 90% power to detect a 10% (54 per 1000 patients per year) between-arm difference in the overall rate of antibiotic prescribing (two-sided $\alpha=0.05$), assuming a correlation of 0.82 between baseline and outcome antibiotic prescribing rate and analysing using analysis of covariance (ANCOVA) with baseline prescribing rate as covariate and 5% practice drop-out rate. For 80% power, 47 practices are needed in each arm. Sample size calculations were performed in PASS 2019 software.

Given the limitation on numbers in the BRIT2 intervention arm, our primary target is to recruit 112 practices although we will attempt to recruit up to 124 practices. However, determination of the total sample size of practices will depend on practice consent to take part in the trial and passing an assessment of IT system capability for the KSS installation.

Note: The definition of 'practice' was changed slightly post-randomisation. As some of the units randomised worked closely together and merged after randomisation, they were classed post-hoc as a practice, and any such 'practice' was allocated to the intervention arm unless all the organisational units forming the newly defined 'practice' had been allocated to the enhanced-care (control) arm, in which case the newly-defined practice was allocated to the enhanced-care arm.

2.4 Framework

All outcomes will be tested for superiority of the BRIT2 Intervention over Enhanced care (i.e. KSS+prescriber feedback dashboards over standard care+prescriber feedback dashboards).

2.5 Statistical interim analysis and stopping guidance

No interim analysis is planned.

2.6 Timing of final analysis

All outcomes analysed collectively at the time of final data *identification and extraction*. This will be (approximately) June 2025 once all (or almost all) events occurring during the trial period up to end May 2025 can reasonably be assumed to have been entered into the Graphnet systems. Analyses will take place once the final trial dataset has been formed from the data extracted from Graphnet and other sources and approved by the Statisticians and Chief Investigator.

2.7 Timing of outcome assessments

The trial will use 1st May 2024 as the start point for the study. Outcomes shall be assessed based on data collected during the 12 months from 1st May 2024 to 30th April 2025, except for infection-related complications for which the collection of data will extend 30 days beyond this date (i.e. to 30th May 2025).

3 STATISTICAL PRINCIPLES

3.1 Confidence intervals (CI) and level of statistical significance

Analyses will be conducted at the 5% significance level and by using two-sided 95% confidence intervals. No formal adjustments will be made for multiplicity as there is a single primary analysis. Interpretation of the tests for significance of the secondary outcomes will made only if the primary analysis demonstrates statistical significance ($p<0.05$).

3.2 Adherence and protocol deviations

3.2.1 Adherence

Adherence is defined as the number of KSS activations divided by the total number of initial GP consultations for common infections in adult patients during the trial period. Whenever a clinician activates the KSS, this usage will be logged with time stamps along with information on which KSS pages were accessed (no patient-level data are stored in this log file). Adherence will be summarised as follows:

$$\text{adherence} = \frac{\text{number of KSS activations}}{\text{number of initial consultations for common infections}}.$$

Due to the novelty of the BRIT2 intervention, a suitable level of adherence cannot be estimated prior to analysis. Potential cut-offs for 'acceptable' adherence will be examined in an exploratory (post-hoc) manner where levels of adherence across all practices will determine the cut-offs used for the analyses. For example, the lowest cut-off will be >0, where any practices that use the KSS is classed as adherent, and other cut-offs may be, for example 25%, 50% and 75% adherence.

3.2.2 Non-compliances (Protocol deviations)

Major protocol deviations include:

1. Any practice allocated to the BRIT2 intervention arm for which the KSS system was not installed before 1st May 2024.
2. GP practices merged post-randomisation if group allocation was affected by the merging (i.e. if any merged practices were formed from at least one practice allocated to the 'Enhanced Care' intervention group and at least one practice allocated to the 'BRIT2' intervention group, the merging of the practices will be classed as having caused a major protocol deviation).

Minor protocol deviations include any technical issue that would prevent the KSS from being used as intended which would lead the GP to consult and/or prescribe the antibiotic in a different manner:

1. System failures or access issues which prevent the KSS from being used during consultation.

3.3 Analysis populations

Data from each participating practice will be anonymised patient-level data using electronic health records. Patient-level data will be aggregated at the practice level. This practice-level data will then be used for the statistical analyses described within this SAP.

The analysis population from both the baseline year and the trial period will be a complete-case population (i.e. excluding any practices that withdraw during the trial period and do not provide consent to use their data, or for whom data are or become unavailable for research). Patients in participating practices who have opted out of NHS data sharing for secondary use will not be included in the aggregated data set prepared for analyses, nor will patients who have left the practice prior to the final data extraction.

Per-protocol analysis populations will also be formed to exclude practices:

(i) with technical non-compliance; and
(ii) which were merged post-randomisation.

as defined (as major protocol deviations) in section 3.2.2.

The safety outcome population will comprise all registered adult patients from each GP practice who have an initial consultation for a common infection during the trial period.

4 TRIAL POPULATION

4.1 Screening data

Practices will go through a screening process to ensure they are suitable for the KSS arm which includes IT service provider support and computer system requirements. Basic information on practices who were screened but not eligible to enter the trial will be collected and summarised.

4.2 Eligibility

Inclusion criteria:

- Any General Practice where it is (deemed) possible to install the KSS will be eligible to take part.
- Patients at least 18 years of age (as the KSS can only be activated for adults).

Exclusion criteria:

- Patients who have opted out of NHS data sharing.
- Patients who left a study practice (including those who died) prior to final data extraction.

4.3 Recruitment

A CONSORT flow diagram showing the number of practices screened, eligible, declined to participate, randomised and withdrawn will be used to summarise the recruitment of practices. A template for the CONSORT flow diagram is provided as Appendix A.

4.4 Withdrawal/follow-up

Practices that withdraw may still provide consent to allow their data to be used. Any practices that withdraw from the study and do not provide consent to use their data will be summarised and excluded from all other analyses.

4.5 Baseline patient characteristics

Wherever appropriate within the analysis, the baseline year is taken as 1st February 2023 to 31st January 2024. The baseline year will be applied to the following characteristics: the mean baseline rate of AB prescribing, consultation rates for common infections, and rates of infection-related complications.

Practice size will be summarised as appropriate, using the mean (SD) and median (IQR), for both the overall number of registered patients and numbers of registered adult patients, based on the final data extraction. This number of registered adult patients will also be used as the denominator for rates for the baseline characteristics detailed below (as the corresponding numbers during the baseline year were not obtainable).

Baseline practice characteristics will be summarised at the practice level both overall and by treatment arm. These will be summarised using mean (SD) (if continuous and symmetrical), and/or the median and interquartile range (if not symmetrical or not expected to be relatively symmetrical). Categorical variables at the practice level will be summarised by frequencies and percentages, or as the mean (SD) and median (IQR) percentage, as appropriate. Statistical tests will not be conducted to test for differences between arms.

Baseline characteristics will be summarised at the practice level, and will include:

- Baseline rate of antibiotic prescribing (i.e. the mean (SD) and median (IQR) rate of antibiotic prescriptions across all practices during the baseline year). The antibiotic prescribing rate within each practice will be 'per 1000 eligible registered adult patients per year'.
- Mean proportion (SD) and median (IQR) for 'low' or high' rates of AB prescribing across all practices (stratification factor). The median baseline rate of AB prescribing (46.64 AB prescriptions per 1000 registered patients per year) was used to dichotomise all participating practices as 'low' or 'high' prescribers. Practice AB rates used in randomisation were determined as described in section 2.2.
- Region (stratification factor)
    - Greater Manchester
    - Wirral
    - Merseyside/Cheshire.
- Mean (SD) age in years of adult patients across all practices using the mean age within each practice. The age for each patient will be determined as the age at the start of the trial period (i.e. 1st May 2024).

- Mean (SD) and median (IQR) proportion of *adult* patients who are at least 65 years old at the start of the trial period.
- Mean (SD) and median (IQR) proportion of patients who are at least 65 years old at the start of the trial period
- Mean proportion of adult male patients across all practices using the proportion of adult males within each practice. Patients within each practice where gender was 'undisclosed' will be excluded.
- Mean (SD) and median (IQR) Index of Multiple Deprivation (IMD) rank across all practices. The median IMD ranking (from IMD 2019 [7]) for adult patients within each practice will be used as the IMD for that practice.
- Mean (SD) and median (IQR) proportion of adult patients for each IMD quintile across all practices. IMD quintiles within each practice will be determined as the proportion of adult patients in each category of: most deprived, deprived, medium, affluent, most affluent. Categories will be derived using the rankings and deciles from IMD 2019 [7].
- Mean (SD) and median (IQR) proportion of adult patients for each UK ethnic group category across all practices. Ethnic group distribution within each practice will be determined as the proportion of adult patients in each ethnicity category of: White, Asian, Black, Mixed, Other, Unknown (this category will combine adult patients where ethnicity is either 'undisclosed' or missing).
- Mean (SD) Charlson Comorbidity Index (CCI) of adult patients with a common infection across all practices using the mean CCI within each practice [8].
- Mean (SD) and median (IQR) proportion of adult patients for each index category of the CCI across all practices. The CCI category 'score' within each practice will be determined as the proportion of adult patients in each of the following five CCI categories: Very low (CCI=0), Low (CCI=1), Medium (CCI= 2), High (CCI=3 or 4), Very high (CCI=5 or more)).
- Mean (SD) and median (IQR) proportion of patients who are at least 65 years old with a moderate or severe frailty score (i.e. frailty score > 0.24 [9]).

  (Note: This proportion of patients within each practice will be calculated using the number of patients aged over 65 years old as the denominator, and the number of patients aged over 65 and with a frailty score > 0.24 as the numerator.)

- Mean (SD) and median (IQR) consultation rate across all practices. Consultation rates for common infections within each practice (i.e. the rate of clinical codes for any common infection during the baseline year) will be per 1000 adult registered patients per year where the common infections are:

    - UTI
    - Cough
    - Sore throat
    - Otitis media
    - Otitis externa
    - Sinusitis.
- Mean (SD) and median (IQR) rate of infection-related complications. Rate of infection-related complications within each practice will be per 1000 registered adult patients.
- Mean (SD) and body mass index score (BMI) across all practices. Mean BMI score of adult patients within each practice will use the most recent BMI record.
- Mean (SD) and median (IQR) proportion of adult patients for each category of smoking status across all practices. Smoking status within each practice will be determined as the proportion of adult patients in each category of: never smoked, ex-smoker, current smoker and unknown, using the most recent smoking status record.
- Characteristics of adult patients with **common infections** during the baseline year.
    - Mean (SD) age (years) of adult patients using the mean age within each practice.
    - Mean (SD) proportion of adult male patients across all practices using the proportion of adult males within each practice.

5 EFFECTIVENESS ANALYSIS

5.1 Outcome definitions

5.1.1 Primary outcome

The primary outcome will be the overall rate of antibiotic prescribing to adult patients (defined as aged 18 or over on 1st May 2024) during trial period from 1st May 2024 to 30th April 2025. Antibiotic prescribing rates will be determined for each practice and then averaged across all practices to calculate the overall rate for each trial arm (i.e. BRIT2 intervention arm; enhanced care arm). Antibiotic prescribing rates will be calculated using the number of AB prescriptions per 1,000 registered adult patients per year.

Note: The type of infection may not be recorded by GP practices and so the overall rate of prescribing will include *all* indications. If records from GP practices show more than one antibiotic prescription for the same indication/infection, these will be counted as separate prescriptions.

5.1.2 Secondary outcomes

a. Rate of infection-related complications (as defined by Gulliford et al [4]) within 30 days of initial GP consultation by an adult patient (aged 18 or over on 1st May 2024) such as pneumonia and lower RTIs, peritonsillar abscess, mastoiditis, intracranial such abscess, empyema, scarlet fever, pyelonephritis, septic arthritis, osteomyelitis, meningitis, toxic shock syndrome and septicaemia, and Lemierre syndrome as recorded in the primary care records, for a common infection. Where relevant, this will include data up to 30th May 2025 to account for common infections during the final month of the trial period. If records from GP practices show repeat consultations within 14 days of each other for the same infection, the first GP consultation will be counted as day 1 of the 30-day period. Records which show a consultation for multiple common infections on the same day will be excluded. If there are consultations for multiple common infections within 14 days of each other, then only the first record for a consultation will be used. Complications recorded on the same day of the consultation will also be excluded. Rate of infection-related complications will be the number of complications per 1000 registered adult patients per year. [Safety outcome]

b. Risk-based prescribing for adult patients prescribed an antibiotic at an initial consultation for a common infection during the trial period (i.e. the risk of hospitalisation due to an infection-related complication in the 30 days after the initial consultation with associated antibiotic prescription). For each antibiotic prescription on the day of a consultation for a common infection, the predicted risk of hospitalisation due to an infection-related complication will be estimated based on the validated risk prediction score for that common infection [5]. For each practice, the predicted risks will be averaged for all adult patients prescribed an antibiotic at an initial consultation for a common infection to form the main secondary outcome variable.

   Note: It is expected that any risk-based prescribing by GPs would lead to an increased mean predicted risk in the intervention arm (than in the control arm) as antibiotic prescribing should be more targeted to those with higher risk.

5.2 Analysis methods

5.2.1 Primary outcome analysis

The primary analysis will be performed at practice level using ANCOVA, weighed by practice size (number of registered *adult* patients), and **with adjustment for practice-level baseline variables**:

- Antibiotic prescribing rate during the baseline year;

- Region;
- Medan patient IMD rank;
- Mean age of adult patients;
- Proportion of adult male patients;
- Proportion of adult patients from each UK ethnic group category;
- Consultation rates for any common infection during the baseline year.

The primary analysis will be a complete-case analysis (i.e. excluding practices who explicitly withdraw their consent to allow their data to be used).

5.2.2 Secondary outcome analysis

The analysis of the secondary outcomes will use a similar approach to that of the primary outcome: ANCOVA with adjustment for the same set of practice-level baseline variables, except that the baseline level of antibiotic prescribing used to stratify the randomisation will be included as the simple dichotomous stratification factor, and adjustment for the mean age and proportion of males for adult patients with common infections during the baseline year (rather than for all adult patients).

In addition, the ANCOVA models will be adjusted for the following practice-level variables:

- Mean CCI within each practice for adult patients who experience any common infection during the trial period; and
- For the model for the rate of infected-related complications only, the baseline level of infection-related complications.

5.2.3 Sensitivity analyses

Sensitivity analyses for the primary outcome will include per-protocol analyses for which the respective per-protocol populations (see section 3.3) will be formed by:

1. Excluding practices that merged after randomisation to different trial arms.
2. Excluding practices that did not have the KSS installed before 1st May 2024.

A sensitivity analysis for the secondary outcome variable ‘Rate of infection-related complications’ will also be performed. This analysis will use the same statistical model as detailed in section 5.2.2 but using an alternative definition of that outcome. In this definition, ‘per 1000 infections (recorded as initial consultations for common infections) per year amongst adult patients’ will be used to compute the rate, rather than ‘per 1000 registered adult patients per year’.

5.2.4 Subgroup analyses

Subgroup analyses for the primary outcome will consist of modelling of the primary outcome by practice-level subgroups. Subgroups are:

- region (Greater Manchester; Wirral; Merseyside/Cheshire) [stratification factor];
- baseline antibiotic prescribing level (high; low) [stratification factor];
- age (above or below median proportion of adult patients who are at least 65 years across study practices);
- socioeconomic level (above or below median England and Wales IMD rank [of 16,422]);
- frail elderly (above or below median practice proportion of patients aged at least 65 yrs who have a frailty score > 0.24 [9]).

Each subgroup analyses will be implemented by adding an appropriate interaction between that categorical variable and the trial arm factor (BRIT2 or enhanced care arm).

5.3 Missing data

Since statistical analyses will be performed at the practice-level, we do not anticipate missing values at practice level, and so imputation methods will not be applied. Missing data is likely to occur at patient level for characteristics such as ethnicity and smoking status, and we will account for missing information or ‘undisclosed’ by including a category for ‘unknown’. Any potential missing information at patient level for other characteristics (e.g. age and sex) are expected to be few (or none) and will not have an important impact on the analyses.

5.4 Additional analyses

1. A CACE analysis of the primary outcome using randomisation as an instrumental variable, adjusting for adherence of the KSS in the BRIT2 intervention arm.
2. The usage of periodic feedback dashboard systems during the trial period will be analysed. The rate of instances dashboards was accessed during the trial period (per 1000 patient-years) will be determined for each practice, and the mean rate will be summarised across all practices and compared descriptively between trial arms.
3. Trends in usage of the KSS will be examined at monthly intervals across the trial period. Rates of KSS usage (i.e. the number of activations per 1000 registered adult patients per year) will be summarised across all practices and a plot produced to show the trend of rates throughout the trial period at monthly intervals.

5.5 Harms

Infection-related complications within 30 days of initial GP consultation for a common infection as defined in Section 5.1.2.

5.6 Statistical software

Statistical analyses at the practice level will be performed using Stata.

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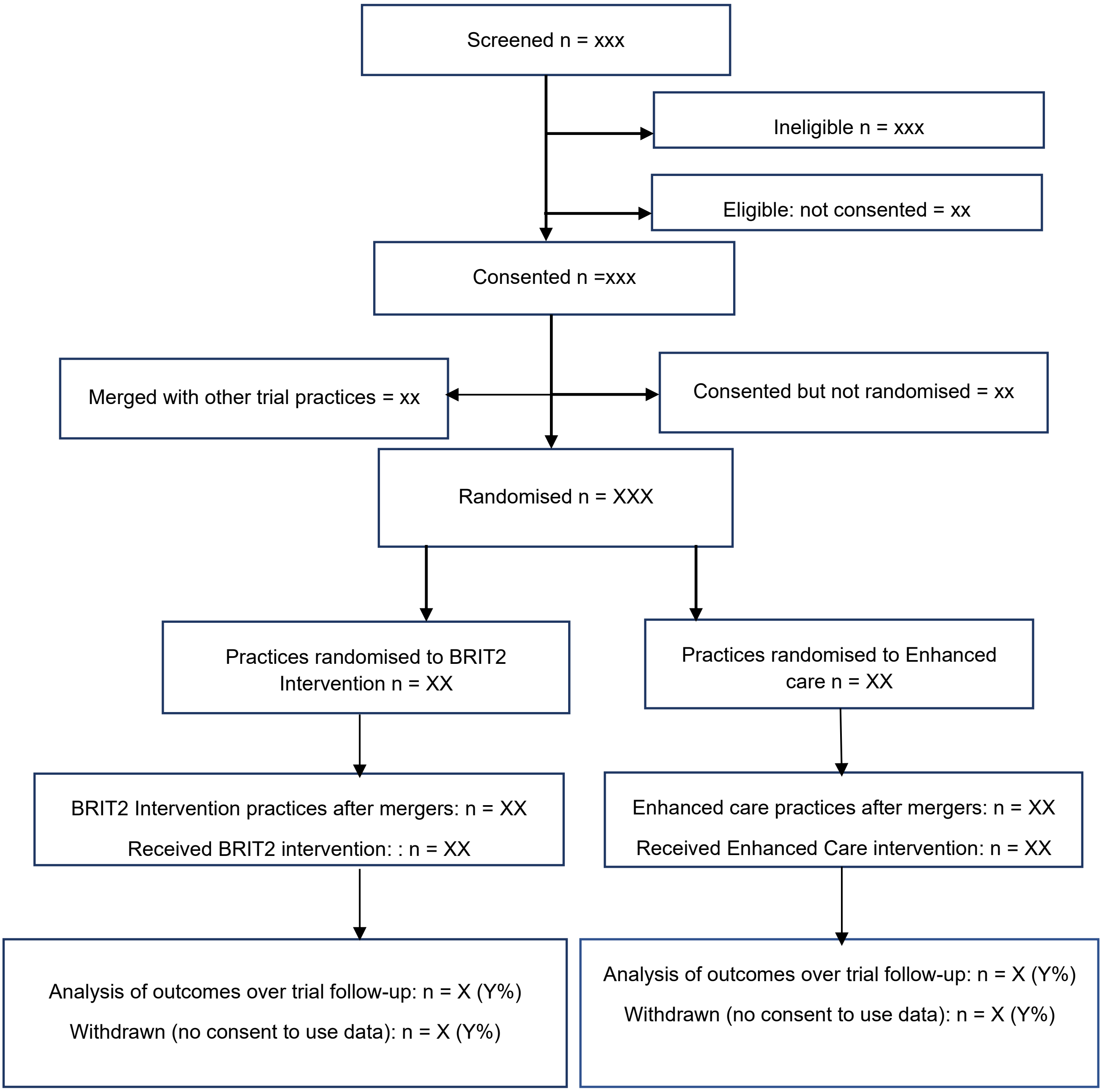
7 APPENDICES

7.1 Appendix A: CONSORT flow diagram template

BRIT2 CONSORT diagram – recruitment and follow-up of practices



7.2 Appendix B: Variables included in the risk-based prescribing models

Hospital admission due to infection-related complications [5]

Variables	Type	Categories
Age	Categorical	18-24, 25-34, 35-44, 45-54, 55-64, 65-74, 75 or more
Sex	Categorical	Male, Female
Ethnicity	Categorical	White, Non-white, Unknown
Region	Categorical	Northwest
Smoking	Categorical	Smoker, Ex-smoker, Never smoked, Unknown
Body mass index category	Categorical	Underweight (>10 and ≤18.5), Healthy weight (>18.5 and ≤25), Overweight (>25 and ≤30), Obese (>30)
IMD quintile **	Categorical	1-5 (1=Most deprived to 5=Most affluent)
Flu vaccination in the last 1 year	Categorical	Yes, No
CCI category ***	Categorical	Very low (=0), Low (=1), Medium (=2), High (=3 or 4), Very high (=5 or more)
Season	Categorical	Spring, Summer, Autumn, Winter
Count of prescribed antibiotics in the last 1 year	Numerical	-

** IMD, Index of Multiple Deprivation.

*** CCI, Charlson Comorbidities Index, measured from 17 weighted conditions, including myocardial infarction, congestive heart failure, peripheral vascular disease, cerebrovascular disease, dementia, chronic pulmonary disease, Connective tissue disease, ulcer disease, mild liver disease, diabetes, hemiplegia, moderate or severe renal disease, diabetes with complications, any malignancy (including leukaemia and lymphoma), moderate or severe liver disease, metastatic solid tumour, and AIDS.

Outcome

The outcome is the risk of hospital admission related to incident infections, which includes cough, sore throat, urinary tract infection, sinusitis, otitis media, and otitis externa. The risk is a value between 0 and 1.

7.3 Appendix C: List of infection-related complications that led to hospital admissions due to a common infection

Source: OpenCodelists: Hospital admissions with infection-related complication

ICD-10	Infection-related complication
A37	Whooping cough
A38	Scarlet fever
A39	Meningococcal infection
A40	Streptococcal sepsis
A41	Other sepsis
A46	Erysipelas
A48	Other bacterial diseases, not elsewhere classified
A49	Bacterial infection of unspecified site
B95	Streptococcus and staphylococcus as the cause of diseases classified to other chapters
B96	Other specified bacterial agents as the cause of diseases classified to other chapters
G00	Bacterial meningitis, not elsewhere classified
G01	Meningitis in bacterial diseases classified elsewhere
H60	Otitis externa
H65	Nonsuppurative otitis media
H66	Suppurative and unspecified otitis media
H67	Otitis media in diseases classified elsewhere

H68	Eustachian salpingitis and obstruction
H700	Acute mastoiditis
J00	Acute nasopharyngitis [common cold]
J01	Acute sinusitis
J02	Acute pharyngitis
J03	Acute tonsillitis
J04	Acute laryngitis and tracheitis
J05	Acute obstructive laryngitis [croup] and epiglottitis
J06	Acute upper respiratory infections of multiple and unspecified sites
J13	Pneumonia due to Streptococcus pneumoniae
J14	Pneumonia due to Haemophilus influenzae
J15	Bacterial pneumonia, not elsewhere classified
J16	Pneumonia due to other infectious organisms, not elsewhere classified
J17	Pneumonia in diseases classified elsewhere
J18	Pneumonia, organism unspecified
J20	Acute bronchitis
J21	Acute bronchiolitis
J22	Unspecified acute lower respiratory infection
J36	Peritonsillar abscess
J85	Abscess of lung and mediastinum
J86	Pyothorax
K05	Gingivitis and periodontal diseases
K61	Abscess of anal and rectal regions
L00	Staphylococcal scalded skin syndrome
L01	Impetigo

L02	Cutaneous abscess, furuncle and carbuncle
L03	Cellulitis
L04	Acute lymphadenitis
L05	Pilonidal cyst
L08	Other local infections of skin and subcutaneous tissue
M00	Pyogenic arthritis
M86	Osteomyelitis
N10	Acute tubulo-interstitial nephritis
N30	Cystitis
N390	Urinary tract infection, site not specified
N45	Orchitis and epididymitis
N70	Salpingitis and oophoritis
N71	Inflammatory disease of uterus, except cervix
N72	Inflammatory disease of cervix uteri
N73	Other female pelvic inflammatory diseases
N74	Female pelvic inflammatory disorders in diseases classified elsewhere
N75	Diseases of Bartholin gland